Test Coverage Report

Document control

Forever Hotel • Manager Dashboard • Guide deliverable 6

Prepared: 2026-10-25, Asia/Colombo. Milestone: approximately 80% project progress, as stated by the owner. Evidence status: Coverage not measured for this snapshot.

Source: development, caef8822afc0c44b030bea2453e23eb01d4af42a. Prepared for student/team review; author identity and approval are not provided. The guide-specified filename includes “sprint8”; no completed Sprint 8 is asserted.

1. Executive assessment

Unit, API/database integration, component and BFF test sources exist for the MAD implementation. CI is configured to collect and enforce coverage. However, no current backend/frontend coverage artifact or successful run tied to the selected commit was available locally. Consequently this report cannot certify achieved line/branch coverage or the guide's endpoint-coverage gate.

No tests were executed to prepare this PDF. Source inspection is reported as source inspection; missing measurements are not converted into pass percentages.

2. Interpretation of the 80% milestone

The included scope is the MAD code already built: manager authentication, reporting, staff administration, promotions, complaints, operational snapshots/events, notifications, exports, settings and supporting gateway transport.

The remaining work includes current execution/coverage evidence, provider integration acceptance, measured performance, resolution of intermittent 503s, production security/deployment verification and manager/supervisor UAT. No numerical weighting of those work packages has been established. This report does not multiply coverage by 0.8, select convenient passing tests or omit failures.

3. Guide requirements and actual evidence

Guide section 6.4 defines three distinct requirements. The owner's project-progress estimate does not lower or replace them.

Requirement	Required threshold	Current evidence	Assessment
New-code unit lines	At least 80%	Configured in backend/ frontend; no measured artifact	Not measured

Requirement	Required threshold	Current evidence	Assessment
New-code unit branches	At least 80%	Configured global coverage gates	Not measured
Critical business logic branches	At least 90%	Selected critical module and changed-branch gates	Not measured
API endpoint integration coverage	Every endpoint has at least one integration test	Integration sources and route-family mapping exist; no exhaustive executed endpoint audit	Not established
CI-generated report linked to sprint review	Dated report/run linked to reviewed commit	No matching run URL or review evidence attached	Pending

4. Static test-source inventory

Counts are test files, not executed cases or passed tests. A unit spec may contain many cases, and an integration suite may use fixtures/reference owners rather than a real remote provider.

Test source group	Files found	Source pattern
Backend unit	38	backend/src//*.spec.ts
Backend API/DB integration	16	backend/test/*.e2e-spec.ts
Frontend unit/component/ BFF	34	frontend/test//*.test.{ts,tsx}
Gateway fixture test source	1	api-gateway/test/*.test.ts

The current frontend package exposes Vitest tests, not an active Playwright test:e2e script. Archived browser logs are not current browser acceptance. The gateway is a separate repository; its coverage cannot be silently combined with MAD into one achieved percentage.

5. Measurement configuration and scope

Backend Jest uses V8 coverage. Its configuration includes source TypeScript with specific exclusions and per-file thresholds; consult backend/package.json for the exact current coverage selectors. Tests marked .e2e-spec.ts use backend/test/jest-e2e.json and disposable resources.

Frontend Vitest uses V8 with text, LCOV and JSON-summary reporters. Its selected scope includes libraries, components, proxy, specified API route groups and the room page; lib/navigation.ts is excluded. It is not literally every frontend page. Source: frontend/vitest.config.ts.

The changed-coverage script examines covered changed lines/branches against a base SHA, including stricter branches for selected critical code. Passing global coverage alone does not prove those gates pass, nor does it prove test quality. Source: scripts/check-changed-coverage.mjs.

6. Results ledger

Metric	Backend	Frontend	Gateway
Executed cases passed / failed	Not available	Not available	Not available
Lines covered / total	Not measured	Not measured	Not measured
Line coverage percentage	Not measured	Not measured	Not measured
Branches covered / total	Not measured	Not measured	Not measured
Branch coverage percentage	Not measured	Not measured	Not measured
Critical-branch gate	Not verified	Not verified	No identical MAD gate claimed
Endpoint integration percentage	Not established	BFF source cases exist; not measured	Not established
CI run and execution date	Not supplied	Not supplied	Not supplied

There is no evidence-backed pass, fail or measured coverage score for the chosen commit in this report. An unrun gate is pending, not a passing gate.

7. Endpoint coverage review map

This identifies relevant integration sources. It is a planning map rather than proof that every method, action, alias, denial and error branch has been executed.

Route family	Integration source	Review still needed
/auth/*, /health/*	backend/test/foundation.e2e-spec.ts and unit auth specs	Confirm every auth action and health alias under the final configuration
/mad/analytics/bookings, revenue, overview	bookings.e2e-spec.ts, report/ foundation suites	Exact controller-to-case audit for overview and revenue
Occupancy / low-booking	occupancy.e2e-spec.ts	Data completeness, all invalid ranges and live 503 cause

Route family	Integration source	Review still needed
Food / services / workers	food-analytics.e2e-spec.ts, service-request-analytics.e2e-spec.ts, worker-performance.e2e-spec.ts	Boundary/status/data-quality cases at the final commit
Staff	staff.e2e-spec.ts	Each role/create/read/update/reset action and privilege rejection
Promotions	promotions.e2e-spec.ts	Every lifecycle action, room types, publication and version conflict
Complaints	complaints.e2e-spec.ts	List/detail/transitions/private history and grants
Kitchen / ticket / sockets	kitchen-realtime.e2e-spec.ts	Real producer and browser timing are separate acceptance work
Rooms / tasks	rooms.e2e-spec.ts, tasks.e2e-spec.ts	Mutations intentionally rejected; aliases/filter denials
Notifications / stock	notifications.e2e-spec.ts, stock.e2e-spec.ts	Read-state and source clearing/duplicate cases
Settings task limit	task-limit.e2e-spec.ts	Actual WKMS owner, not only the reference fixture
Reports	reports.e2e-spec.ts	Every report type, format, size limit and filter combination
Operations	Unit backend/src/operations/operations.spec.ts	Endpoint integration proof not established by this map

All filenames without a prefix in this table are under backend/test/.

8. Reproducible execution and artifact handoff

Use approved CI and disposable test services; never run integration fixtures on the hosted application database. Provide the required TEST_DATABASE_URL and TEST_RABBITMQ_URL securely. Do not include their values in the report.

Backend directory: npm ci

Backend directory: npm run test:cov -- --runInBand

Backend directory: npm run test:integration

Frontend directory: npm ci --ignore-scripts

Frontend directory: npm run test:cov

Gateway directory: npm ci --ignore-scripts

Gateway directory: npm run test:cov

CI also runs lint, formatting, build/config integrity and dependency audit checks. Current MAD CI has no deployment, browser acceptance or container-smoke stage. Attach the run URL, tested SHA, tool/runtime versions, execution date, source selectors, LCOV/summary output and test result counts. Upload artifacts before copying actual values into section 6. Resolve real gaps rather than excluding difficult files simply to reach a threshold.

9. Milestone completion gate

The coverage PDF is prepared as a status artifact for an 80%-complete project. The coverage deliverable's passing-results gate remains pending. It can be marked accepted only after measured coverage, endpoint review and matching CI evidence are attached and reviewed. This report does not certify 80% measured coverage.

References

Guide sections 6.3.3, 6.4, 7 and 10.1; backend/package.json; frontend/vitest.config.ts; .github/workflows/ci.yml; scripts/verify-quality-gates.mjs; scripts/check-changed-coverage.mjs; documents/testing/test-register.md; local original SRS/SDS under docs/requirements/.